Structural Equation Modelling (SEM)

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1 Introduction

SEM is a diverse set of statistical tools that are generally applied in social and behavioural sciences. SEM is so diverse that can be a course in itself and is heavily used in studying consumer behaviour. Therefore, we will only focus the basic concepts of SEM and how can they applied in different situations. Regression models that you have learnt in the previous lectures also fall under SEM amongst other models such as confirmatory factor analysis (CFA), path analysis etc.

In multiple linear regression you have already studied how to associate multiple independent variables with an dependent variables, and to infer if the associations between the independent and the dependent variable are positive or negative or non-existent based the β's of the independent variable. In other words, you learnt in linear regression, how much variance of the dependent is explained by the independent variables using the R^2 value. In SEM we go a couple to steps further to study more complicated but still **linear** relationships!

SEM consists of two parts: (1) Structural model and (2) Measurement model.

Hence the name, Structural Equation Modelling (SEM) = Structural model + Measurement model. We will look into each of these part through an example in the next section.

2 Basic concept

Let us look into an example from (Streiner, Finding our way: an introduction to path analysis, 2005), where the author studies a phobia called ***photonumerophobia*** which *is the fear that our fear of numbers will come to light*. Yeah, you read it right! The fear that our fear of numbers will be exposed. Anyway moving ahead, the author hypothesized that photonumerophobia (PNP) can be predicted using 3 factors:

1. overall level of anxiety (ANX),
2. high school math grade (HSM), and
3. the discrepancy between what the person estimated last year's taxes to be and what the actual tax was (TAX).

Multiple regression: $\boldsymbol{PNP = \beta_0 + \beta_1 ANX + \beta_2 HSM + \beta_3 TAX + \epsilon}$

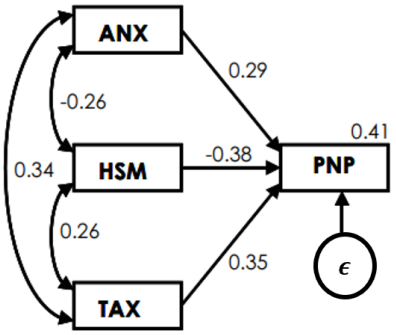


Figure 1: Simple multiple linear regression model as formulated to study relationship between three factors anxiety (ANX), *high school math grade (HSM), and tax discrepancy (TAX) with photonumerophobia (PNP) (Streiner, Finding our way: an introduction to path analysis, 2005).*

Initially a simple multiple linear regression model was formulated as shown in Figure 1. The double headed curved arrows between ANX-HSM, HSM-TAX, and ANX-TAX are the correlations between the independent variables. You can see that ANX and HSM have a negative correlation. The single headed arrow shows the relationship of the independent variables with the dependent variable, e.g., ANX -> PNP imply, ANX helps in explaining PNP through β_1. The strength of the relationship depends on the magnitude and the sign of β_1. In Figure 1 $\beta_0 = 0.41,\ \beta_1 = 0.29,\ \beta_2 = -0.38,\ \beta_3 = 0.35$, and ϵ is the normally distributed error term. By now, you know how to interpret these variables from your multiple regression lecture.

2.1 Structural model (aka path analysis)

Until now you haven't learnt anything new, but it's time. What the author found out that the R^2 of the model in Figure 1 was rather low and they hypothesized two new models as shown in Figure 2:

- In Figure 2 (a), it's hypothesized that, rather than the variables acting separately on PNP, there is truly a **path**, with anxiety (ANX) leading to poor math grades in high school (HSG), which results in mistakes in completing tax forms (TAX) and culminates in PNP.
- In Figure 2 (b), the hypothesis is that anxiety results in both poor grades and computational errors in filling out taxes and that both of these together lead to PNP.

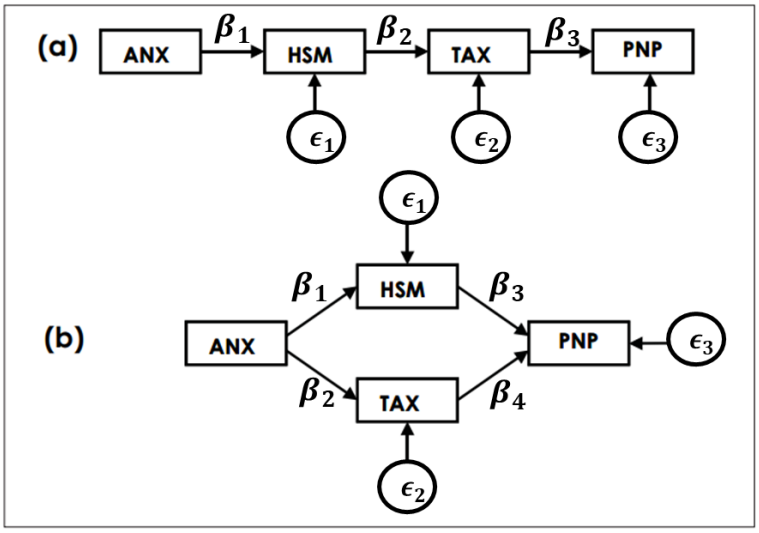


Figure 2: Two different hypothesized structural (path) models for explaining PNP through ANX, HSM, and TAX by *(Streiner, Finding our way: an introduction to path analysis, 2005)*.

More formally we say that the effect of ANX on PNP is **mediated** by HSM and TAX through the **structure** or **path** hypothesized in Figure 2. Hence the name **structural model** or **path analysis**.

Structural model (**path analysis**): *This part of the SEM model defines the relationship between the variables. It specifies the manner by which particular variables directly or indirectly influence (i.e., "cause") changes in the values of certain other variables in the model* (Byrne., 2013).

Note that in the structural model:

- The β's are identical to the weights in multiple linear regression.
- The β's should be statistically significant (remember the p-value).
- There are several fit indices that tell you how well the model fits the data, we will them discuss later.

2.2 Measurement model (aka factor analysis)

The *measurement model* involves variables that cannot be measured directly (**latent variables**) and are generally inferred using other measurable (**observed**) variables.

Let us look into the anxiety (ANX) variable discussed in Section 2.1. When we say a person is anxious, what we mean is that he or she is showing *observable behaviours* that are manifestations of one or more of these facets. In the physiological realm, for instance, there may be tachycardia, shortness of breath, and sweatiness (all of which are *measurable*). In the cognitive realm, there may be decreased ability to concentrate and hypersensitivity to perceived threats (again *measurable*). In other words, anxiety is something we cannot directly measure, but can be hypothesize to tie together observable phenomena that are correlated with each other to some degree. Therefore, anxiety is a **latent** variable which is measure through some **observed** variable such as heart rate, poor concentration etc. You can think of the latent variables as *hidden* variables.

In a subsequent paper (Streiner, Building a better model: an introduction to structural equation modelling., 2006) formulated that anxiety has 4 facets—cognitive, affective, behavioural, and physiological—that are themselves unobservable hypothetical constructs. The measurement model for anxiety is shown in Figure 3.

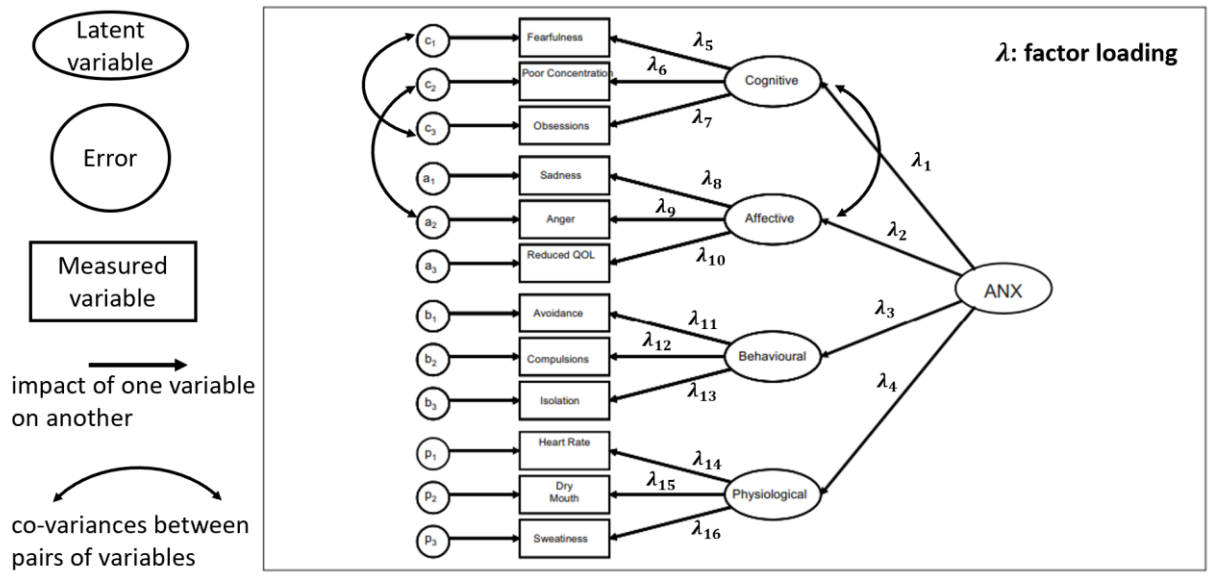


Figure 3: Measurement model for anxiety as described in *(Streiner, Building a better model: an introduction to structural equation modelling., 2006)*.

There is lot of information in Figure 3, let us go through them one by one:

- **Oval**: The latent variables are always represented with an oval shape. In Figure 3, ANX, cognitive, affective, behavioural and physiological are the latent variables.
- **Rectangle**: The observed variables (which can be directly measured) are always represented with a rectangle shape. The cognitive latent variable is measured through three observed variable fearfulness, poor concentration and obsession. For details look into the paper how the authors come this conclusion.
- **Circles**: these are the errors associated with the measured/observed variables. Note these are sample data, and are bound to be corrupted with errors. Every measurement has an error associated with it. These error terms are the unexplained variances for each of the variables.

- **Double headed curved arrows**: these are covariances between two variables, applicable for both the latent and the measured (observed) variables. Which two variables will be correlated is defined by the researcher/modeller. So it will be your educated choice, which variables you think are correlated to each other, ideally should be based on some theory. In Figure 3, cognitive is correlated with affective; fearfulness is correlated with obsession; and poor concentration is correlated with anger.
- **Single headed arrows:** these are the impact of one variable on the other. You might be confused why doesn't the single headed goes from cognitive -> ANX, instead in Figure 3 it is the opposite, it goes from ANX -> cognitive. This is because a person's cognitive ability is affected due to anxiety and not the other way round! Similarly if someone has physiological issues then their heart rate will be affected and not the other way round. Therefore the arrow goes from physiological -> heart rate.
- **Loading factors $\boldsymbol{\lambda's}$:** The $\lambda's$ (lambda's) on the single headed arrows are called *factor loadings*. These can be seen as $\beta's$ as in multiple regression model. The $\lambda's$ should be statistically significant (remember the p-value). In Figure 3, ANX has an influence of λ_1 on cognitive; cognitive has influence on fearfulness, poor concentration and obsession of λ_5, λ_6 and λ_7 respectively. The interpretation of the $\lambda's$ are similar to the interpretation of $\beta's$ in multiple regression model.

Generally for the measurement model, researchers have a theory and conceptualization of the latent variables. Then they design measurement instruments such as a *questionnaire* or *test* that provides them with multiple indicators of the latent variable. The elements of the measurement instruments form the measured/observed variables.

2.3 SEM model specifications

In Section 2.1 you learnt how more complicated linear models to can be formulated to define relationships between variables in structural model. In Section 2.2 you learnt how latent (hidden) variables can be measured using observed variables in the measurement model. As mentioned in Section 1 that SEM is formed by adding these two models: (1) structural model and (2) measurement model together.

For the example we have looked into so far, the SEM model for predicting a subject's degree of PNP on the basis of ANX, HSM, and TAX is modelled as shown in Figure 4 (Streiner, Building a better model: an introduction to structural equation modelling., 2006).

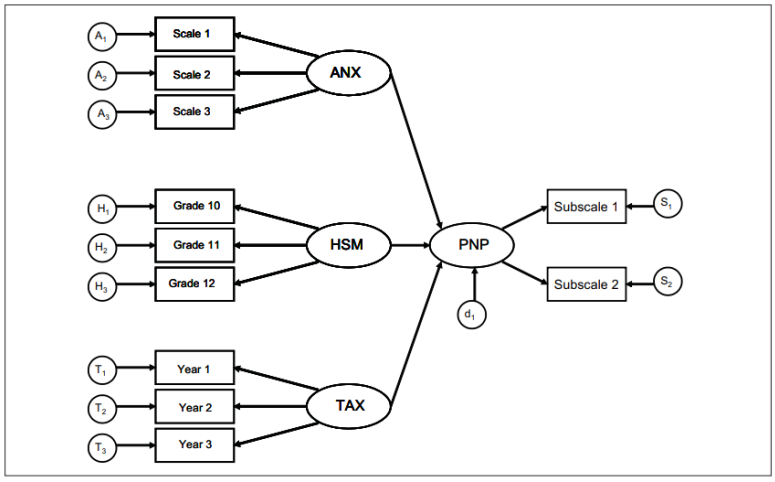


Figure 4: SEM model for predicting a subject's degree of PNP on the basis of ANX, HSM, and TAX *(Streiner, Building a better model: an introduction to structural equation modelling., 2006)*

In Figure 4, you can see that even the PNP is a latent variable measured through two subscale. These will most likely be a questionnaire created by the researchers which are given to a sample of individuals to fill. Note that the structural models here is a simple model where the variables ANX, HSM and TAX acting separately on PNP, notice the direction of the one headed arrows here. They are the same as in Figure 1. For the SEM model in Figure 4, the researchers did not add any correlated variables.

Few things to remember while specifying a SEM model:

- The first step in SEM is called the *model specification* stage.
- In this step we specify the model variables and it is probably the most difficult and most important part. It is the most difficult step because it requires the most thought and understanding of the theoretical model of the purported influences on the variable of interest.
- No computer program can help us at this stage, only our knowledge of the field. It is the most important step because everything depends on how well we specify the model.
- The computer programs may help us in determining whether some variables aren't important, we can play with different *structures*/*paths* to see whether they improve the model.
- The primary cause of poorly fitting models is the omission of crucial variables, and there are no programs in the world that can help us in this regard.
- For example, if the prime determinant of PNP is actually the PNP level of one's parents (because of either genetics or learning), and if this isn't correlated with any of the other variables we're examining, our model will explain little of the variance, and we will never know why.

2.4 SEM model fitting

This step is an easy step. Generally we use a computer program to fit the hypothesized SEM model to the data. This step involved computing all the model parameters $\boldsymbol{\theta} = [\boldsymbol{\lambda}, \boldsymbol{\beta}, \boldsymbol{\Sigma}]$, where $\boldsymbol{\lambda}$ is the vector to all the loading factors, $\boldsymbol{\beta}$ is the vector to all the regression coefficients, and $\boldsymbol{\Sigma}$ is the covariance matrix.

After computing $\boldsymbol{\lambda}$ and $\boldsymbol{\beta}$ one should check if they all have the right sign. Are they significant? If the answer to either question is no, it may be best to respecify the model by dropping nonsignificant variables and (or) seeing whether there are variables in your data set that should be included.

Generally for SEM we use the **maximum likelihood estimation (MLE)** method to estimate the model parameters. We will not go into details of this algorithm as it is beyond the scope of this course. However, in general the parameter estimation is done by comparing the population co-variance matrices representing the relationships between variables and the estimated co-variance matrices of the best fitting model.

2.5 SEM model fit evaluation

After the SEM model is fitted to data, one needs to test the goodness of the model fit, just like in multivariate regression model we check the R^2 value. Generally a researcher will formulate multiple competing SEM models and will evaluate which model fits the data best based on few metrics.

The output SEM programs includes metrics of the estimated relationships between variables in the model. There are differing approaches to assess model fit. Because different assessment of fit capture different elements of the fit of the model, it is appropriate to report a selection of different fit measures. Some of the more commonly used measures of fit include:

- **Chi-squared p-values:** The *null hypothesis* is the difference between the observed covariance matrix and the model co-variance matrix is zero, and *the alternative hypothesis* is the difference is not equal to zero. **Note:** For this test if the $p > \alpha$, the pre-defined significance level, then we fail to reject null and conclude that the model fit is good.

- **Root Mean Square Error of Approximation (RMSEA):** difference between the observed correlation and the predicted correlation. Fit index where a value of zero indicates the best fit. While the guideline for determining a "close fit" using RMSEA is highly contested, ***most researchers concur that an RMSEA of .1 or more indicates poor fit.***

- **Standardized Root Mean Residual (SRMR):** The SRMR is a popular absolute fit indicator and is defined as the standardized difference between the observed correlation and the predicted correlation. ***A value less than .08 for SRMR is generally considered a good fit.***

- **Comparative Fit Index (CFI):** In examining baseline comparisons, the CFI depends in large part on the average size of the correlations in the data. If the average correlation between variables is not high, then the CFI will not be very high. ***CFI value of .95 or higher is desirable.***

- **Akaike information criterion (AIC):** *A test use to compare two competing models***, the *preferred model is the one with the lowest AIC value***. $AIC = 2k - 2\,ln\,(L)$, where k is the number of parameters in the statistical model, and L is the maximized value of the likelihood of the data given the model parameters.

The different metrics have different purpose depending upon the problem at hand. For example, if you want to compare two competing models, first you will choose the model with a lower AIC value and then for the chosen model look at its Chi-squared p-values, RMSEA, SRMR and CFI. Based on all the values you should conclude if the model is a good fit or not.

2.6 SEM example

Let us have a look into another more complex SEM model. The authors in this study integrated leadership into the job demands-resources model and argued that engaging leaders who inspire, strengthen, and connect their followers would reduce employee's levels of burnout and increase their levels of work engagement. They inferred from an online survey of a representative sample of the Dutch workforce ($n = 1213$) that leadership only had an indirect effect on burnout and engagement – via job demands and job resources – but not a direct effect. They also inferred that leadership had a direct relationship with organizational outcomes such as employability, performance, and commitment (Schaufeli, 2015).

From Figure 5, try to recognize the latent variables, the measurement variables, the covariances, the $\lambda's$, the $\beta's$, the structural model and the measurement model.

NOTE: Generally the relationships between the latent and the measurement variables comprise of the measurement model and the relationships between the latent variables are part of the structural model.

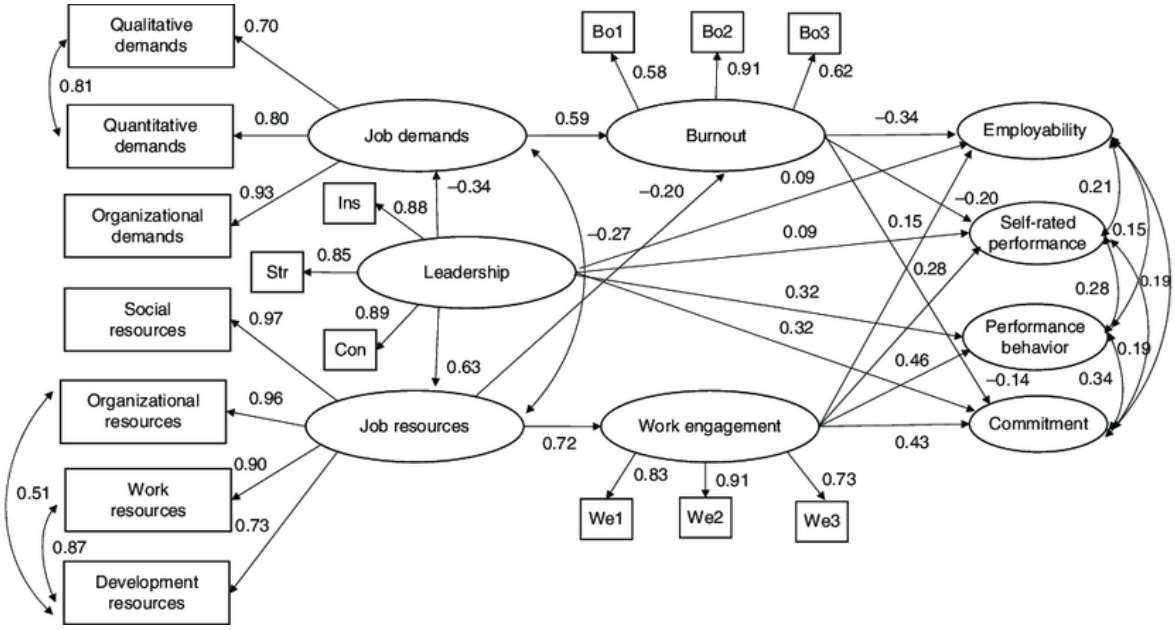


Figure 5: SEM model to study the relationship of leadership and employee burnout *(Schaufeli, 2015)*.

3 SEMOPY python package for SEM:

In the class I will demonstrate how to write a code for SEM for the famous political democracy dataset by Bollen. We will use a Python package called SEMOPY (semopy: Structural Equation Modeling in Python).

You can already install the package on your laptops before coming to the class. Also go through the shared python notebook on Brightspace.

4 Bibliography

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Streiner, D. L. (2005). Finding our way: an introduction to path analysis. *The Canadian Journal of Psychiatry*, 50(2), 115-122.

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